

Signal classification of Non-Standard Interactions at JUNO

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Abstract

In this work, we investigate the detectability of Non-Standard Interactions (NSI) in neutrinos using Machine Learning (ML) techniques. The study was conducted in the context of the Jiangmen Underground Neutrino Observatory (JUNO), considering atmospheric neutrinos, originating from pions decay. The oscillation probability was implemented using the perturbative expansion presented in [1]. Simulated events were generated for Standard Model (SM) and NSI scenarios, incorporating matter effects and experimental resolution. Different MLs were compared, including Random Forest, XGBoost, and Multilayer Perceptron (MLP) neural networks. Furthermore, we investigated the dependence of the classifier's performance on physical and experimental parameters, such as NSI intensity, experimental resolution, and number of events.

1 Introduction

Neutrino physics has become one of the most active areas of modern particle physics following the discovery of neutrino oscillations [2]. Although the Standard Model of three-flavor oscillations successfully describes most current experimental results, several fundamental questions remain open. These include the precise determination of the mass hierarchy and the measurement of the leptonic CP violation phase.

Among the various proposed scenarios, Non-Standard Interactions constitute a particularly interesting possibility. NSIs introduce new effective couplings that can modify neutrino propagation and alter oscillation probabilities [3, 7]. Depending on the intensity and structure of these interactions, the observed effects can mimic or mask fundamental parameters of the standard oscillation model. A major difficulty in the search for NSI stems from the strong degeneracy that exists between new physics parameters and standard parameters, especially involving the leptonic CP violation phase. In many scenarios, different parameter combinations can produce very similar experimental signatures, making the separation between standard events and events containing new physics signals a highly challenging problem.

In this context, next-generation experiments play a central role. The JUNO experiment, currently one of the most important initiatives in neutrino physics, was designed to provide high-precision measurements using a liquid scintillator detector of 20 kton [8]. One of JUNO's most striking features is its excellent energy resolution, designed to reach approximately:

$$\frac{\sigma_E}{E} \approx \frac{3\%}{\sqrt{E(\text{MeV})}}, \quad (1)$$

representing one of the best precisions ever proposed for a large-scale neutrino detector. The experiment was initially conceived for studies involving reactor neutrinos and determination of mass hierarchy, but recent analyses by the collaboration also demonstrate great potential for studies with atmospheric neutrinos. Furthermore, studies indicate sensitivity to atmospheric neutrinos across a wide energy range, between 100 MeV and 10 GeV [9].

However, modern experiments like JUNO will produce extremely large amounts of experimental data. The combination of large statistics, multiple observable variables, and complex oscillatory structures makes traditional analysis based solely on simple cuts or manual adjustments increasingly inefficient.

In this scenario, machine learning techniques are assuming an increasingly important role in high-energy physics. Supervised methods are capable of identifying subtle correlations in multi-dimensional spaces and exploring patterns that are difficult to visualize directly using traditional methods. These techniques become especially relevant when seeking to identify weak or strongly degenerate signals, as expected in scenarios involving NSI.

In this work, we investigate the possibility of distinguishing signals with and without NSI in atmospheric neutrinos using ML techniques in the context of the JUNO experiment. The main objective is to evaluate the separability between standard and NSI scenarios, study the impact of physical and experimental parameters on classification, and investigate the performance of different supervised learning models.

2 Theoretical Formalism

The oscillation probability used in this work was based on the perturbative expansion presented in the article [1]. The transition probability with NSI can be written as:

$$P_{\mu e}^{NSI} = P_{\mu e}^{NSI}(0) + P_{\mu e}^{NSI}(1/2) + P_{\mu e}^{NSI}(1) + P_{\mu e}^{NSI}(3/2) + \dots \quad (2)$$

The dominant terms used in this work are given by:

$$\begin{aligned} P_{\mu e}^{NSI} = & 4 \frac{s_{13}^2 s_{23}^2}{(1 - r_a)^2} \sin^2 \left(\frac{(1 - r_a) \Delta L}{2} \right) \\ & + \frac{8 J_r r_\Delta}{r_a (1 - r_a)} \cos \left(\delta_{CP} + \frac{\Delta L}{2} \right) \sin \left(\frac{r_a \Delta L}{2} \right) \sin \left(\frac{(1 - r_a) \Delta L}{2} \right) \\ & + p_\mu^2 |\epsilon_{\mu e}|^2 \\ & + 4 p_\mu |\epsilon_{\mu e}| \frac{s_{13} s_{23}}{1 - r_a} \sin \left(\frac{(1 - r_a) \Delta L}{2} \right) \sin \left(\delta_{CP} - \phi_{\mu e} + \frac{(1 - r_a) \Delta L}{2} \right) \end{aligned} \quad (3)$$

Where s_{ij} and c_{ij} represent $\sin \theta_{ij}$ and $\cos \theta_{ij}$ for the standard PMNS mixing angles [4]. The parameter Δ is defined as $\Delta \equiv \frac{\Delta m_{31}^2}{2E}$. L is the distance between the source and the detector. The ratio between the solar and atmospheric mass scales is expressed by $r_\Delta \equiv \frac{\Delta m_{21}^2}{\Delta m_{31}^2}$. The matter potential induced by terrestrial electron density is entered through the variable $r_a \equiv \frac{a}{\Delta}$, where $a = \sqrt{2} G_F N_e$ [5]. The reduced Jarlskog factor is given by $J_r \equiv c_{12} s_{12} c_{23} s_{23} s_{13}$ [6], while δ_{CP} represents the Dirac phase. The coefficient p_μ is the chiral factor derived via quantum field theory for the pion decay in the source [1]. Finally, the parameters $|\epsilon_{\mu e}|$ and $\phi_{\mu e}$ define, respectively, the magnitude and complex phase associated with NSI coupling at the neutrino production vertex.

3 Data Generation and Experimental Simulation

To perform the ML analysis, we generated independent simulated datasets for both the SM and NSI scenarios. The data generation pipeline combines analytical physics computations via Monte Carlo sampling with models of detector response and systematic uncertainties. For each scenario, a consistent dataset of $N = 5 \times 10^4$ events was constructed.

3.1 Monte Carlo Event Generation

The input variables for the simulation represent the physical parameters of atmospheric neutrinos originating from pion decays, in the context of JUNO. The kinematics and phase space of the events were sampled according to the following distributions:

- **Neutrino Energy (E):** Sampled from a uniform distribution in the range $E \in [0.2, 10]$ GeV, encompassing the relevant spectrum for atmospheric neutrino oscillations at JUNO [9].
- **Baseline (L):** Sampled from a log-uniform distribution spanning $L \in [20, 12700]$ km.
- **CP-violating Phase (δ_{CP}):** Sampled uniformly between 0 and 2π .

The standard three-flavor neutrino oscillation parameters were fixed throughout the simulation in accordance with [1]. The specific values utilized, including the Earth matter density, are summarized in Table 1.

Table 1: Fixed oscillation parameters and matter density used in the simulation.

Parameter	Fixed Value
$\sin^2 \theta_{12}$	0.307
$\sin^2 \theta_{13}$	0.022
$\sin^2 \theta_{23}$	0.56
Δm_{21}^2	$7.53 \times 10^{-5} \text{ eV}^2$
Δm_{31}^2	$2.51 \times 10^{-3} \text{ eV}^2$
ρ	2.5 g/cm^3

To evaluate the impact of matter density variations along the atmospheric neutrino path through the Earth, a comparative check was performed using two different profiles: a central mantle density ($\rho = 5.0 \text{ g/cm}^3$) and a crust/mantle density ($\rho = 2.5 \text{ g/cm}^3$). The results indicate that shifting the matter density introduces only minor quantitative modifications in the oscillation probability, concentrated primarily at low energies. Within the scope and core objectives of this project, these variations do not yield any significant impact on the final classification performance, thereby justifying the adoption of a constant average density profile for the general analysis.

3.2 Experimental Resolution and Smearing

A realistic sensitivity study requires accounting for the finite performance of the experimental apparatus. To mimetize the reconstruction capabilities of the JUNO detector and introduce conservative systematic bounds, a Gaussian smearing procedure was applied to both the energy and the computed probabilities.

The reconstructed energy E_{rec} is modeled by fluctuating the true Monte Carlo energy E using a Gaussian random variable approach. Specifically, E_{rec} is sampled from a normal distribution centered at E , denoted by

$$E_{\text{rec}} \sim \mathcal{N}(E, \sigma_E^2), \quad (4)$$

where $\mathcal{N}$ represents the normal (Gaussian) distribution parameterized by the mean E and variance σ_E^2 . The energy resolution is parameterized as $\sigma_E = 0.03 \cdot E$, meaning it scales linearly with the interaction energy, where both E and σ_E are explicitly expressed in units of GeV.

Similarly, the experimental uncertainty in the oscillation probability determination, which arises from tracking and angular reconstruction constraints, is implemented by adding a Gaussian noise term to the theoretical probability. This effect is governed by the dimensionless parameter $\sigma_P = 0.001$, defining a fluctuation where

$$P_{\text{smearred}} \sim \mathcal{N}(P_{\text{analytical}}, \sigma_P^2), \quad (5)$$

with $\mathcal{N}$ again denoting the normal distribution. This explicit parameterization ensures that the simulated dataset properly reproduces the standard finite resolution effects expected in the detector response.

3.3 Oscillation Probability

Figure 1 presents the comparison between SM and NSI curves in the probability vs energy plane. For this analysis, the adopted parameters were $L = 1000$ km, $|\epsilon_{\mu e}| = 2 \times 10^{-3}$, $\delta_{CP} = 1.2\pi$ and $\phi_{\mu e} = \pi/2$.

The analysis of the behavior of the probability curves reveals three fundamental physical aspects that directly correlate the graphical behavior with the algebraic structure of the perturbative expansion 3:

- In the low-energy region, the probability amplitude in the NSI scenario is noticeably lower than that of the standard model. Numerically, this behavior is governed by the fourth term (T_4) of the equation 3, whose structure establishes that:

$$T_4 \propto \sin \left(\delta_{CP} - \phi_{\mu e} + \frac{(1 - r_a)\Delta L}{2} \right) \quad (6)$$

If we look at the first oscillation maximum, for example, and at the first term (T_1) of equation 3, we see that the oscillation maximum condition establishes that the factor $[(1 - r_a)\Delta L/2]$ is approximately $\pi/2$. At this point, the behavior of the term (T_4) becomes mostly dependent on the relationship between the two phases through the trigonometric identity:

$$\sin \left(\delta_{CP} - \phi_{\mu e} + \frac{\pi}{2} \right) = \cos(\delta_{CP} - \phi_{\mu e}) \quad (7)$$

This relationship dictates a clear mathematical rule for peak behavior:

- The amplitude increases (Constructive Interference): This occurs if $\cos(\delta_{CP} - \phi_{\mu e}) > 0$, which happens when the difference between the phases is in the first or fourth quadrant of the trigonometric circle. The term (T_4) becomes positive, adding to the standard probability.
- The amplitude decreases (Destructive Interference): This occurs if $\cos(\delta_{CP} - \phi_{\mu e}) < 0$, a scenario in which the phase difference is located in the second or third quadrant. The term (T_4) assumes a negative value, subtracting from the total probability. In the specific case of Fig. (1), the adopted phases ($\delta_{CP} = 1.2\pi$ and $\phi_{\mu e} = 0.5\pi$) result in a relative difference of $\Delta\Phi = 0.7\pi$ (126°) and we have ($\cos(126^\circ) \approx -0.588$). Consequently, the term (T_4) acts by subtracting from the standard probability, justifying the flatness and smaller amplitude of the curve with NSI.
- Intersection point and equivalence scenario. An exact intersection point is observed between the two curves. At this specific point, the overall net effect of new physics is completely nullified. From an analytical point of view, this point maps the solution of the equation $T_3 + T_4 = 0$, where the purely NSI term (T_3), strictly positive and constant, is exactly canceled by the interference term ($T_4 = -T_3$), resulting in the equality:

$$\sin \left(\frac{(1 - r_a)\Delta L}{2} \right) \sin \left(\delta_{CP} - \phi_{\mu e} + \frac{(1 - r_a)\Delta L}{2} \right) = -\frac{p_\mu |\epsilon_{\mu e}| (1 - r_a)}{4s_{13}s_{23}} \quad (8)$$

Given that the right-hand side of the equation is a small quantity (~ -0.012), in order for the product of these two sines on the left-hand side to result in a small negative number, there are three different ways this can happen mathematically:

- The value of the first sine is close to zero, then $[(1 - r_a)\Delta L/2] \approx \pi$.

- The value of the second sine is close to zero and the value of the first sine is at its maximum, $[(1 - r_a)\Delta L/2] \approx \pi/2$.
- The value of both sines is close to zero.

The exact position of this intersection is highly sensitive to the choice of phases δ_{CP} and $\phi_{\mu e}$. Phenomenologically, the point where this intersection occurs represents a zone of undetectability or ambiguity, where an experiment operating strictly at this energy would be unable to discriminate the presence of NSI in the source.

- Asymptotic behavior at high energies: For the higher energy regime, the oscillation length grows significantly relative to the distance traveled. Under this condition, all standard and interference oscillatory functions undergo strong kinematic suppression.

$$\lim_{E \rightarrow \infty} T_1 = 0, \quad \lim_{E \rightarrow \infty} T_2 = 0, \quad \lim_{E \rightarrow \infty} T_4 = 0 \quad (9)$$

As a consequence, the standard probability decays asymptotically towards zero. In contrast, the curve with NSI stabilizes at a constant non-zero level, determined exclusively by the boundary term independent of spatial evolution:

$$T_3 = (p_\mu |\epsilon_{\mu e}|)^2 \approx (27 \times 0.002)^2 \approx 0.0029 \quad (10)$$

This behavior demonstrates that the NSI signature at the source is independent of the dynamics of oscillation evolution in vacuum or matter, in the asymptotic regime. Therefore, the high-energy regime constitutes an excellent experimental window to isolate and constrain parameters of new physics, cleanly mitigating the degeneracies and parameter ambiguities that affect the low-energy region.

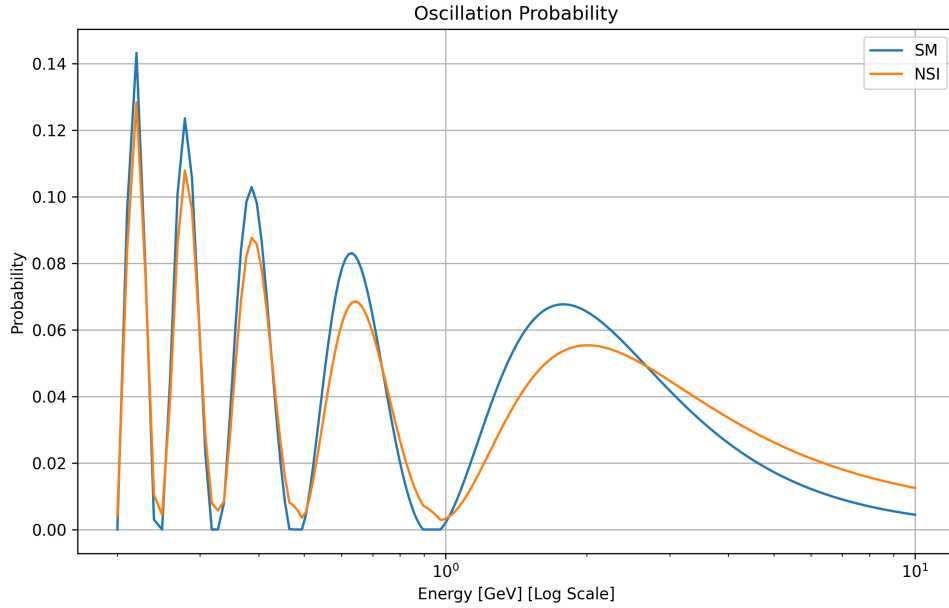


Figure 1: Oscillation probability as a function of energy for SM and NSI scenarios. The mixing parameters were assumed as in Table 1. The others adopted parameters are $L = 1000$ km, $|\epsilon_{\mu e}| = 2 \times 10^{-3}$, $\delta_{CP} = 1.2\pi$ and $\phi_{\mu e} = \pi/2$.

3.4 Dependence on $\epsilon_{\mu e}$

Figure 2 shows the comparison between the oscillation probabilities as a function of energy for the standard scenario and for different values of $\epsilon_{\mu e}$. For this analysis, the parameters $L = 1000$ km, $\delta_{CP} = 1.2\pi$ and $\phi_{\mu e} = \pi/2$ were fixed.

It is observed that as the value of $\epsilon_{\mu e}$ increases, the contribution of the terms associated with NSI becomes more significant, altering both the amplitude and the shape of the oscillations. Furthermore, it is noticeable that the curves exhibit very similar behavior for values in the range of $0 \leq \epsilon_{\mu e} \leq 2 \times 10^{-3}$, while the curve with $\epsilon_{\mu e} = 1 \times 10^{-2}$ stands out from the others.

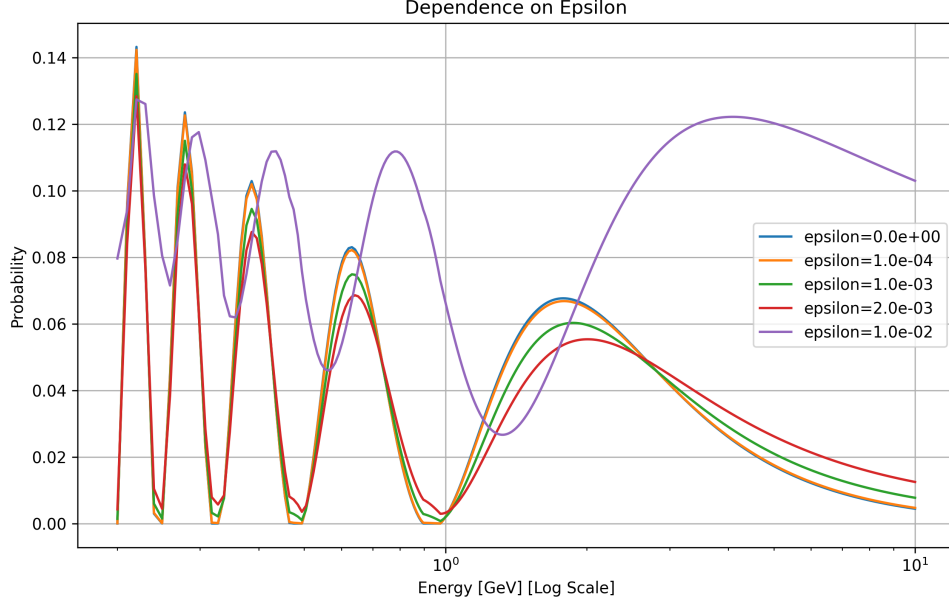


Figure 2: Dependence of the oscillation probability on the NSI parameter $\epsilon_{\mu e}$. The mixing parameters were assumed as in Table 1. The others adopted parameters are $L = 1000$ km, $\delta_{CP} = 1.2\pi$ and $\phi_{\mu e} = \pi/2$.

3.5 Dependence in the NSI phase

Figure 3 shows the dependence of the oscillation probability as a function of the phase associated with the NSI, $\phi_{\mu e}$, keeping the other system parameters fixed. For this analysis, the parameters $L = 1000$ km, $|\epsilon_{\mu e}| = 2 \times 10^{-3}$, $\delta_{CP} = 1.2\pi$ were adopted.

The probability profile reveals a clear regime transition in the grouping of curves along the energy spectrum, which is mostly governed by the dynamic evolution of the interference term (T_4). Analyzing the fifth oscillation maximum, for example, it is possible to note that the oscillation amplitude for the values $\phi_{\mu e} = \pi$ and 1.5π is almost double the amplitude for the values $\phi_{\mu e} = 0$ and 0.5π , showing a high sensitivity to this parameter.

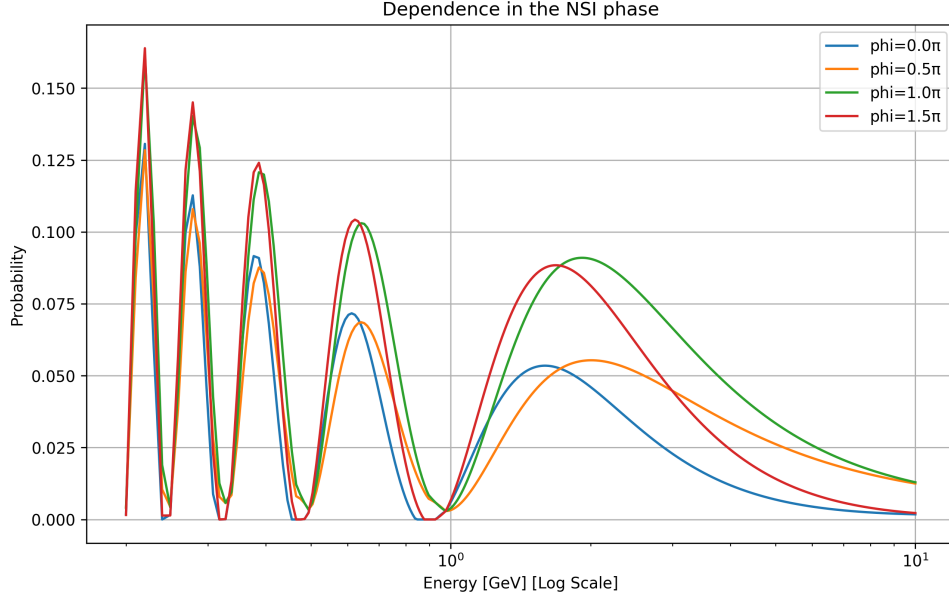


Figure 3: Dependence of the oscillation probability on the NSI phase $\phi_{\mu e}$. The mixing parameters were assumed as in Table 1. The others adopted parameters are $L = 1000$ km, $|\epsilon_{\mu e}| = 2 \times 10^{-3}$ and $\delta_{CP} = 1.2\pi$.

3.6 NSI Probability Heatmap

Figure 4 shows the dependence of the oscillation probability as a function of both the neutrino energy and the baseline distance. For this analysis, the grid parameters were set to a linear distribution of $E \in [0.1, 10]$ GeV and $L \in [20, 12700]$ km, each sampled with 200 points. The variables were fixed with a CP-violating phase of $\delta_{CP} = 1.2\pi$, an NSI coupling magnitude of $|\epsilon_{\mu e}| = 1 \times 10^{-3}$, and an NSI phase of $\phi_{\mu e} = 1.5\pi$.

The plot displays a distinct pattern of alternating interference fringes, which correspond directly to the oscillation maxima and minima of the system. Because the underlying physical mechanism depends strongly on the L/E ratio, these bands form a characteristic diagonal behavior where higher energies require longer baselines to undergo full spatial evolution.

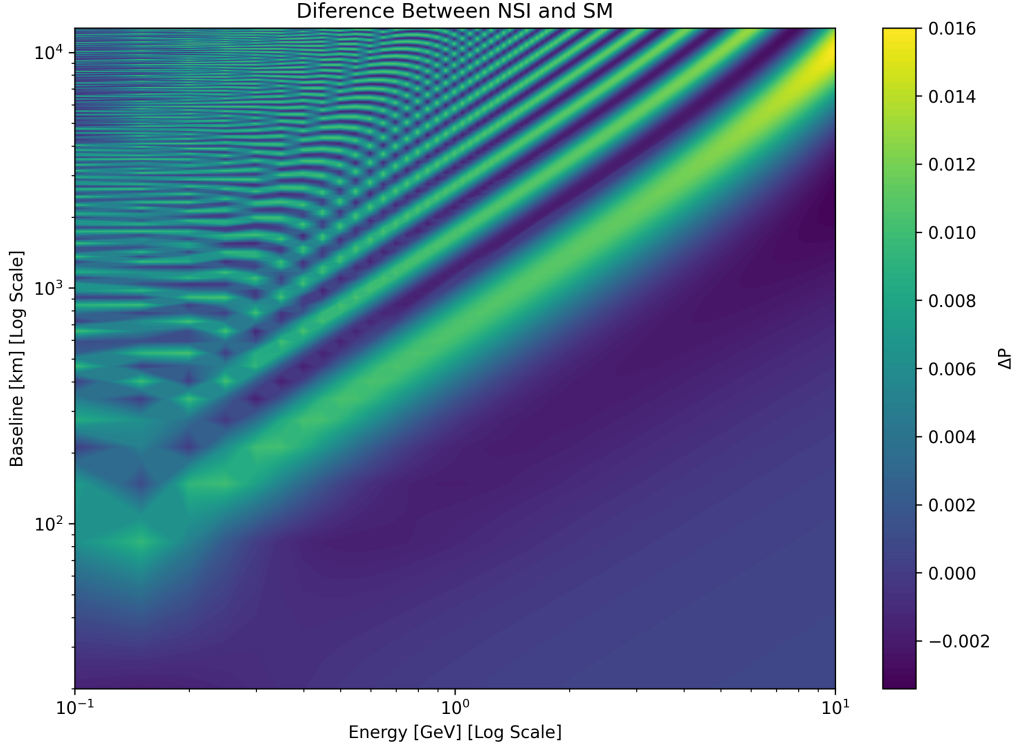


Figure 4: Difference between NSI and SM probabilities. The mixing parameters were assumed as in Table 1. The others adopted parameters are $|\epsilon_{\mu e}| = 1 \times 10^{-3}$, $\delta_{CP} = 1.2\pi$ and $\phi_{\mu e} = 1.5\pi$.

4 Machine Learning

Three supervised classifiers were studied: Random Forest, XGBoost, and a Multilayer Perceptron (MLP) neural network. The original dataset was divided in a 70% training set and a 30% test set ratio.

To mitigate the scale disparity between attributes and ensure stable convergence of the models (particularly the MLP), the features were normalized via StandardScaler. This procedure applies a linear transformation that results in attributes with zero mean ($\mu = 0$) and unit variance ($\sigma^2 = 1$), such that $z = (x - \mu)/\sigma$. The jupyter notebook with the complete code can be found here [13].

4.1 Random Forest

The Random Forest algorithm was adopted as the baseline model. It is an ensemble learning method based on the bootstrap aggregating principle. The model builds multiple independent decision trees in parallel, introducing randomness in both the sampling of training data and the selection of attributes at each split node. The classifier's final prediction is obtained through the majority vote (mode) of the individual trees, which significantly reduces the model's variance and the risk of overfitting compared to a single decision tree.

4.2 XGBoost

The XGBoost (Extreme Gradient Boosting) is an optimized implementation of the Gradient Boosting algorithm that belongs to the class of sequential ensemble methods. Unlike Random Forest, decision trees are built iteratively, where each new tree is trained to minimize the residuals (errors) calculated by the previous trees. The algorithm optimizes a differentiable loss function using a second-order Taylor expansion and incorporates an explicit regularization term ($\mathcal{L}_1$ and

$\mathcal{L}_2$) in the objective function to control the model's complexity. This architecture gives XGBoost high computational efficiency and a great capacity to capture complex and subtle nonlinear relationships in the data.

4.3 Neural Network MLP

A Multilayer Perceptron (MLP) neural network is a deep feedforward learning architecture composed of an input layer, one or more hidden layers with multiple neurons (dense layers), and an output layer. Each neuron calculates the weighted sum of its inputs plus a bias term, subsequently applying a non-linear activation function.

In this work, the hidden layers utilize the Rectified Linear Unit (ReLU) activation function, defined by $f(x) = \max(0, x)$, which solves the vanishing gradient problem and accelerates the convergence of the backpropagation algorithm. The synaptic weights and biases are adjusted iteratively by minimizing the loss function through gradient calculations.

5 Results

5.1 ROC Curves

To evaluate the classifiers' performance in distinguishing between SM events and NSI scenarios, Receiver Operating Characteristic (ROC) curves were generated, as shown in Figure 5. This method allows observation of the model's sensitivity (True Positive Rate) as a function of the false alarm rate (False Positive Rate) as the decision threshold is varied. In the graph, the dashed diagonal line represents the limit of a purely random classifier, corresponding to an Area Under the Curve (AUC) of 0.5.

The results obtained for the different algorithms, Random Forest, XGBoost, and Multilayer Perceptron (MLP), reveal the intrinsic complexity of this classification. Although the curves are positioned above the random reference line, the AUC values indicate limited separation power. Statistically, values in this range demonstrate that the models, while capturing certain correlations in the data, fail to establish a clear decision boundary.

This limitation should not be attributed solely to the architecture of the models, but rather to the physical nature of the problem. There is a significant degeneracy between the oscillations induced by the SM and the NSI corrections, especially when considering the CP violation phase. Furthermore, the noise introduced smooths the spectral differences between the two scenarios. Therefore, the proximity of the curves to the random line reflects the challenge in isolating signals of new physics that manifest as small perturbations relative to the reference model.

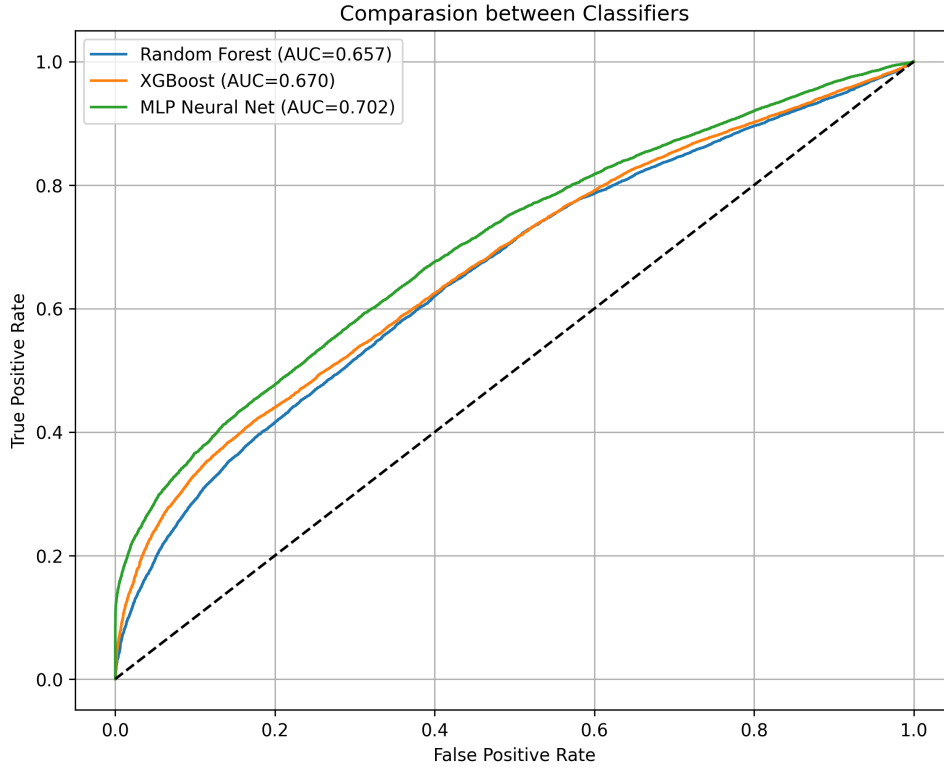


Figure 5: Comparison between the ROC curves of different classifiers.

5.2 Model Performance

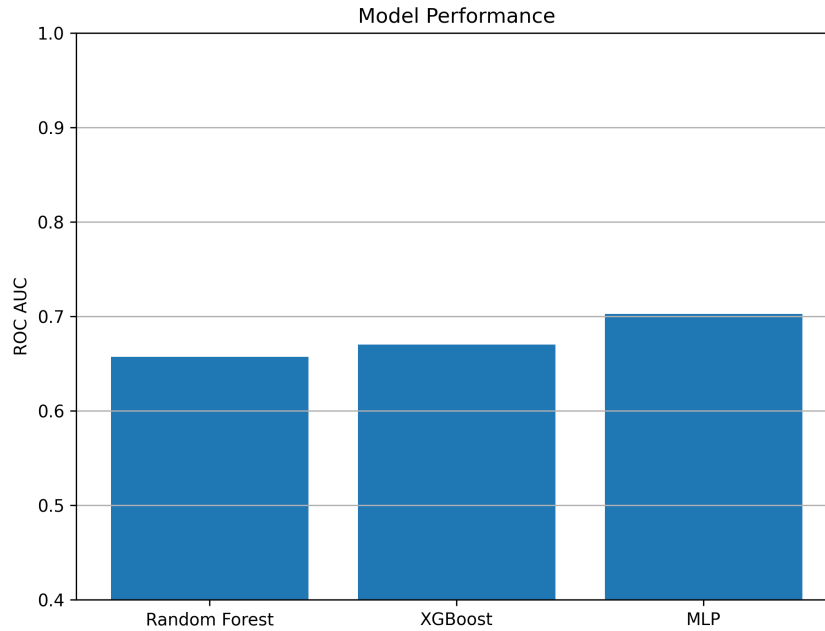


Figure 6: Comparison of AUC for different classifiers.

The comparison of classifier performance, presented in Figure 6, reveals a subtle gradation in the generalization capacity of each algorithm. The Random Forest model presented the lowest AUC value, suggesting a limitation of structures based on averages of independent decision trees for mapping highly smooth transition boundaries in phase space. On the other hand, it is observed

that more sophisticated models, such as XGBoost and the MLP neural network, exhibit better performance.

This divergence in the behavior of the algorithms is directly associated with the ability of each architecture to process complex nonlinear correlations. The XGBoost gradient boosting algorithm, through its sequential residual optimization, and MLP, through its hidden layers and nonlinear activation functions, are able to extract and reconstruct these subtle interference structures and the matter effects generated by the new physics in a slightly more efficient way.

5.3 Importance of Features

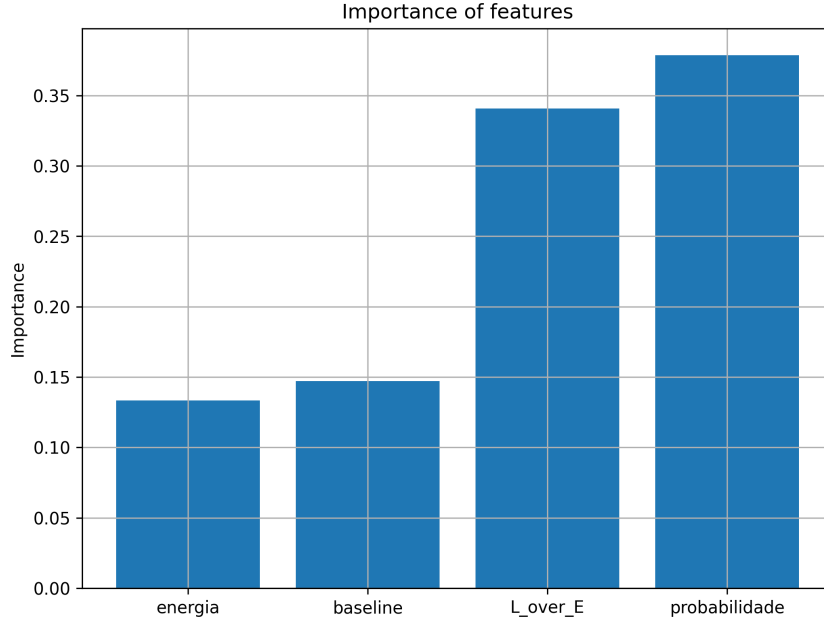


Figure 7: Importance of input variables in Random Forest.

The analysis of the importance of the variables presented in Figure 7 for the Random Forest classifier allows us to identify which kinematic and geometric signatures of the simulated events are crucial for discriminating between the SM and NSI scenarios.

The prominence of probability is a direct consequence of the simulation methodology itself, in which it acts as the main analytical vector that explicitly incorporates the physical signature differentiating the SM and NSI scenarios. Similarly, the high relevance of L/E is justified by the fact that, in propagation dynamics, energy and distance traveled are coupled in this exact functional ratio within the oscillation phases; the classifier, therefore, benefits from the combined variable to map the periodicity of survival profiles more efficiently than through isolated components.

Among the raw kinematic variables, it is observed that L has a slightly higher importance than energy. This asymmetry stems directly from the boundary conditions set during the generation of the dataset. While energy simulates a theoretical uncertainty through the inclusion of Gaussian noise ($\sigma_E = 0.03$), the variable L remains preserved in its pure form, without additional stochastic fluctuations.

5.4 ROC vs $\epsilon_{\mu e}$

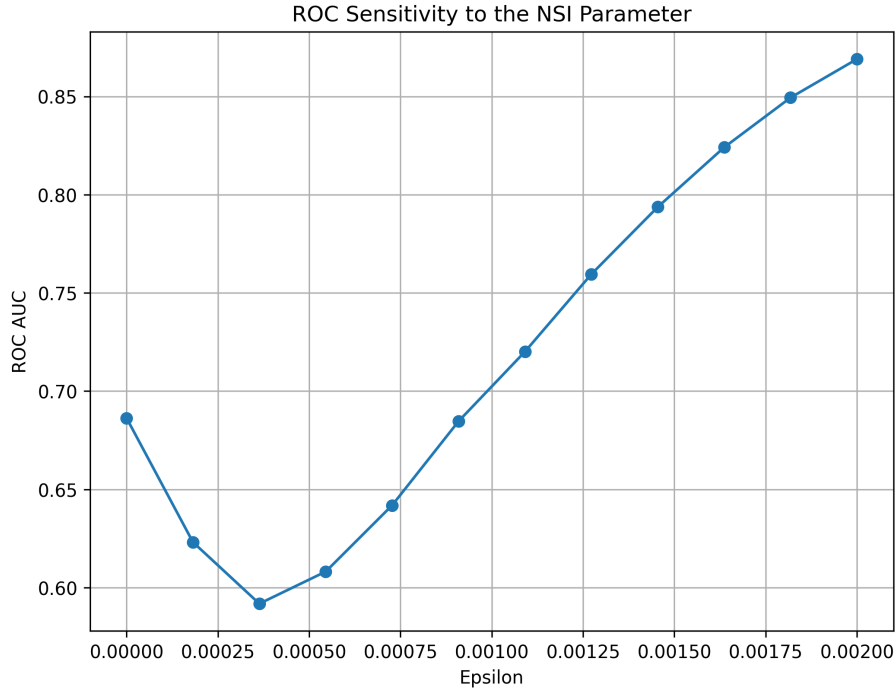


Figure 8: AUC dependency on the NSI parameter $\epsilon_{\mu e}$.

The detectability of the NSI increases with the intensity of the effective coupling. However, non-monotonic behavior is observed for very small values of the parameter, in the region where $0 < |\epsilon_{\mu e}| \lesssim 0.00035$, characterized by a slight initial reduction in AUC.

This inflection stems from the mathematical structure of Eq. 3, where the new physics corrections depend on a linear interference contribution ($\propto |\epsilon_{\mu e}|$) and a purely quadratic contribution ($\propto |\epsilon_{\mu e}|^2$). For very weak couplings, the quadratic term is negligible, and the linear term, subject to constructive and destructive interference along the integration of the phases δ_{CP} and $\phi_{\mu e}$, is masked by the energy smearing and systematic noise introduced in the simulation. The sensitivity assumes a strictly monotonic growth only when the quadratic regime begins to predominate, shifting the probabilities in a definite way and effectively breaking the degeneracy with the Standard Model.

5.5 ROC vs Number of Events

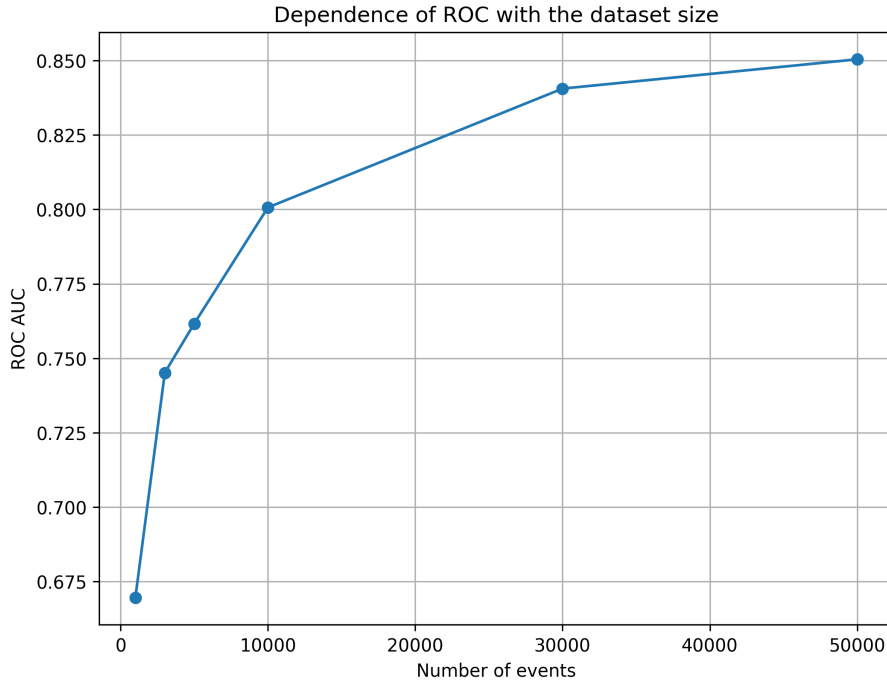


Figure 9: Dependence of AUC on the number of simulated events.

Figure 9 shows the evolution of the AUC metric as a function of the number of simulated events in the dataset. A characteristic asymptotic behavior is observed: the curve exhibits a sharp and steep growth for lower statistics, followed by a deceleration and subsequent stabilization (plateau) at higher event values.

This phenomenon reflects the dynamics of statistical information gain and the fundamental limitations of the classifier. Initially, the error associated with sampling the distributions, which decays with the inverse of the square root of the number of events ($1/\sqrt{N}$), is pronounced. The addition of data rapidly reduces these residual fluctuations, allowing the model to more clearly define the decision boundaries between the SM and NSI scenarios.

However, as the statistics increase, the marginal performance gain saturates. This asymptotic limit indicates that the classifier has reached a performance close to the Bayesian separability limit. Since the model optimally describes the probability densities, the supply of new data is unable to overcome the intrinsic overlap between classes, which is governed by the physical similarities of the oscillation channels and the limitations imposed by the experimental resolutions of energy and detector angle.

6 Discussion

The results indicate that discriminating Non-Standard Interactions in production poses a statistical challenge intrinsic to the physical formalism, not to the algorithms. The moderate AUC values result from the strong parametric degeneracy between the standard CP violation phase (δ_{CP}) and the new physics phase ($\phi_{\mu e}$), as well as the existence of exact intersections in the theoretical probability curves for certain energy values. In these specific regions, the SM and NSI scenarios become mathematically indistinguishable, eliminating any physical signature even before the inclusion of experimental effects.

Despite these physical limitations, models based on non-linear correlations, such as XGBoost and MLP, outperformed Random Forest. The analysis also revealed that the classifier's perfor-

mance is non-monotonic for weak couplings ($|\epsilon_{\mu e}|$), where linear interference terms are masked by systematic uncertainties, becoming steadily increasing only in the regime of dominance of the quadratic term. Finally, the saturation of performance with increasing statistics confirms that the models reached the Bayes separability limit. Under the instrumental conditions tested, providing more data does not overcome the irreducible physical ambiguity of the oscillation channels, validating the use of these techniques to precisely delimit the sensitivity limits of the experiment.

This subtle variation in probability profiles introduces another critical factor: the risk of false positives arising from systematic uncertainties. Since the spectral signature of low-intensity NSI is extremely faint, background statistical fluctuations or inaccuracies in the JUNO detector's energy calibration could easily be interpreted by classifiers as evidence of New Physics, when in fact they only reflect experimental limitations. Therefore, the use of Machine Learning for direct binary classification (SM vs. NSI) proves inadequate.

Given the limitations of direct binary classification, machine learning shows promise when redirected to specific analytical and instrumental tasks within the NSI regime. Firstly, regression models based on deep neural networks can be applied to continuously predict the numerical values of couplings ($|\epsilon_{\mu e}|$) and phases ($\phi_{\mu e}$, δ_{CP}), circumventing the need for a rigid decision boundary and optimizing rapid global adjustments to mitigate degeneracies. At the instrumental level, the use of convolutional neural networks (CNNs) or graph-based neural networks (GNNs) directly on the raw PMT data from the JUNO detector can enhance the reconstruction and calibration of atmospheric neutrinos energy and direction, reducing the smearing effect that smooths the signatures of new physics at low energies. Additionally, classification algorithms can be employed in flavor tagging to discriminate with high precision the final oscillation channels (ν_μ versus ν_e), isolating the spectral regions where NSI effects are prevalent. Finally, the development of classifiers dedicated to rejecting specific backgrounds is fundamental to purging contamination from spurious events, such as reactor neutrinos or geoneutrinos, shielding the sample from real atmospheric neutrinos and maximizing the statistical sensitivity of the experiment.

7 Conclusion

This work investigated the feasibility of using supervised machine learning techniques for discriminating Non-Standard Interactions in atmospheric neutrino production at the JUNO detector. The results demonstrated that binary classifiers face a fundamental separability limit (Bayes limit), imposed by the strong parametric degeneracy between the δ_{CP} and $\phi_{\mu e}$ phases and by the existence of exact intersections in the theoretical probability curves, where the SM and NSI scenarios become indistinguishable. This original physical ambiguity scenario is subsequently amplified by experimental noise, limiting the achievable AUC values.

Although non-linear algorithms such as XGBoost and MLP have outperformed the reference model (Random Forest), the increase in sample statistics confirmed the saturation of performance due to the irreducible overlap of classes. It is concluded, therefore, that formulating the problem as a direct binary classification is not the ideal strategy for the search for New Physics in this context, given the high risk of false positives induced by systematic uncertainties. For future studies, it is recommended to redirect machine learning towards continuous regression tasks aimed at parameter extraction and degeneracy breaking, or towards the instrumental optimization of the detector, including energy calibration, background rejection, and flavor tagging.

References

- [1] Cherchiglia, Adriano and Pasquini, Pedro and Peres, O. L. G. and Rodrigues, F. F. and Rossi, R. R. and Souza, E. S., "Alleviating the present tension between T2K and NO ν A with nonstandard neutrino interactions", *Phys. Rev. D*, 112, 2025.

- [2] Super-Kamiokande Collaboration, "Evidence for Oscillation of Atmospheric Neutrinos", *Phys. Rev. Lett.*, 81, 1998.
- [3] Wolfenstein, L., "Neutrino Oscillations in Matter", *Phys. Rev. D*, 17, 2369–2374, 1978.
- [4] Pontecorvo, B., "Inverse beta processes and nonconservation of lepton charge", *Sov. Phys. JETP*, 7, 1958.
- [5] Wolfenstein, L., "Neutrino oscillations in matter", *Phys. Rev. D*, 17, 1978.
- [6] Jarlskog, C., "Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Violation", *Phys. Rev. Lett.*, 55, 1985.
- [7] Ohlsson, Tommy, "Status of Non-Standard Neutrino Interactions", *Rep. Prog. Phys.*, 76, 044201, 2013.
- [8] An, F. and others (JUNO Collaboration), "Neutrino Physics with JUNO", *J. Phys. G*, 43, 030401, 2016.
- [9] Abusleme, A. and others (JUNO Collaboration), "Atmospheric Neutrinos in JUNO", *JHEP*, 2022.
- [10] P. Huber, M. Blennow, T. Schwetz, and W. Winter, "GLOBES: General Long Baseline Experiment Simulator," *Comput. Phys. Commun.* **167**, 195 (2005), arXiv:hep-ph/0407333.
- [11] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (ACM, New York, NY, USA, 2016), pp. 785–794, arXiv:1603.02754.
- [12] C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, New York, NY, USA, 2006. <https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>.
- [13] <https://github.com/FernandaDeFR/NSI-ML>